

Ageing institutions ossify a cultural field: Evidence from Soviet book publishing for children

Kirill Maslinsky

INALCO

October 7, 2026

Abstract

Some cultural fields undergo an ossification process, with an ever larger share of the produced material featuring already established authors. I argue that this can be explained by the ageing of the institutions of cultural production, without assuming that anyone changed their behavior. The core mechanism is the growing backlog: the probability that a producing unit fills a slot with an item from outside of its own catalog diminishes with the size of that catalog, so a field that stops admitting new producing units will ossify. I test this mechanism on the case of late-Soviet book publishing for children, using digitized bibliographic data covering 1918–1984. Fitting a model in which a board fills a slot with a newcomer at a probability that falls with the size of its accumulated catalog, I find no evidence that the propensity of the boards to publish new authors declined over 1950–1984. Counterfactual tests show that the exposure of an editorial board to its own growing catalog is both necessary and sufficient to explain the diminishing rate of entry of new authors.

Keywords: cultural change, neutral models, institutions, children’s literature, book history

1 Introduction

Some cultural fields undergo what a casual observer may call an ossification process, with an ever larger share of the produced material featuring the already established authors, familiar universes, and recognizable characters. Ossification as a long-running trend has been discussed recently for movies, TV shows, music, video games, and books (Mastroianni 2022). I argue that this effect can be explained by the ageing of the institutions of cultural production, such as publishers and studios, without assuming that anyone changed their behavior.

The growing backlog of published material as the producing institutions age is the core mechanism for the ossification of the repertoire. In the simplest model the probability that a producing unit fills a slot with an item from outside

of its own catalog diminishes with the size of the catalog. When a field stops admitting new producing units, ossification will inevitably ensue. This argument introduces the logic of the neutral-null theory of cultural change at the level of the institutional structure of the field.

In this paper, I test the institutional ageing mechanism on the case of late-Soviet book publishing for children. This case provides a well-defined field with an especially legible administrative history thanks to Soviet bibliographic and publishing institutions. The quantitative study is made possible by the available digitized bibliographic data covering 1918–1984 (Maslinsky 2022).

The data tell a clear ossification story. By the end of the period, the share of printed copies that went to already well-established authors grew five-fold, while the share of editions allocated to new authors fell. In addition, after 1975 new editorial boards almost ceased to appear in the bibliographic record, while the overall number of active boards did not fall.

I fit a model in which a board fills a slot with a newcomer at a probability that falls with the size of its own accumulated catalog, and read off the recruitment intensity per unit of catalog. There is no evidence that the propensity of the boards to publish new authors declined over 1950–1984. Hence the apparent ossification is explained by the ageing population of editorial boards with their accumulated publishing catalogs. Counterfactual tests show that the exposure of the editorial board to its own growing catalog is both necessary and sufficient as a mechanism for the diminishing rate of entry of new authors.

2 Related work

Cumulative advantage is a well-known phenomenon in cultural fields (Merton 1968), and rising concentration is typically taken as evidence about the dispositions of whoever is doing the selecting. In literary studies, the cumulative advantage was explained by the accumulated symbolic capital (Bourdieu 1993), and the existence of specialized institutions that reproduce status and hence participate in the canon formation (Guillory 1993). As was shown in a famous empirical paper, a consumer in a cultural market resolves their uncertainty by relying on the choice of others, and thus success breeds success (Salganik, Dodds, and Watts 2006). But alternative generative mechanisms for the observed frequency distributions are rarely considered (DiPrete and Eirich 2006).

Neutral theory in ecology and evolutionary biology is the tradition that offers such an alternative explanation (Hubbell 2001). Neutral-null models assume that there is no selection pressure, and the cumulative advantage will result from the random choice of variants proportional to their frequency. Neutral models have been applied in cultural evolution studies as well, e.g. for baby name selection (Bentley, Hahn, and Shennan 2004; Kandler and Crema 2019).

The level of analysis that is missing from the cultural literature is the demography of the selecting units themselves. Organizational ecology has studied the founding, ageing and mortality of producing organizations and noted the change in innovative behavior with the organization’s age (Hannan and Free-

man 1984; Sørensen and Stuart 2000). My point is that an ageing population of gatekeepers produces closure with no change in disposition at all. It is what this paper tests.

3 Data

The data are 71,000+ editions of children’s books published in the Soviet Union between 1918 and 1984, one row per edition, carrying the print run (*tirazh*), the year, the publisher, the city, and the author (Maslinsky 2022). The print run is planned supply, fixed in advance by an editorial bureaucracy and printed in the imprint of the book itself. The print run figure is thus the recorded quantity that directly reflects the allocation decision by the editorial board.

The producing unit is the *board* — a publishing house together with its normalized city, which in the Soviet system is an editorial board with its own staff, plan and list. Boards are recovered from the combination of the publisher and city in the imprint, and corrected to take into account mergers and renamings.

For each board b and year t I count the author slots filled, n_{bt} ; how many of them went to an author the board had not printed before, n_{bt}^{entry} ; and the board’s accumulated catalog $M_{bt} = \sum_{s < t} n_{bs}$, its author slots accumulated strictly before t . M is cumulated over the board’s whole observed history. Three filters then apply (Table 1), leaving 4,004 board-years on 571 boards, covering 41,183 author slots.

Panel	board-years	boards	slots
Board-years, 1940–84	4,925	1,041	42,821
+ vintage ≥ 1	4,113	621	41,348
+ $M > 0$ (model panel)	4,004	571	41,183

Table 1: Construction of the analysis panel. Pre-1940 years are dropped because entry counts there are inflated by cataloging recall. A board’s first year (vintage 0) is all entry by construction and so carries no information about recruitment. $M > 0$ is required because the exposure law is undefined at log 0; this drops 109 rows, one per board whose first *authored* year lags its first edition of any kind

The idea that the field of children’s book publishing ossified in the late-Soviet period rests on two trends apparent in the data. First, at the author level the premium on having published before grew with time. The share of printed copies allocated to authors with more than ten prior editions runs from 0.14 in 1925 to 0.78 in 1984, while the share of author slots at a publishing board going to someone that board had never printed before falls from 0.63 in 1940 to 0.44 in 1984. Second, at the field level, the producing population ages and stops turning over. A median of 41 new editorial boards appears in the bibliography per year between 1923 and 1939, against between 2–8 per year after 1975, while 101 to 134 boards went on printing (Figure 1).

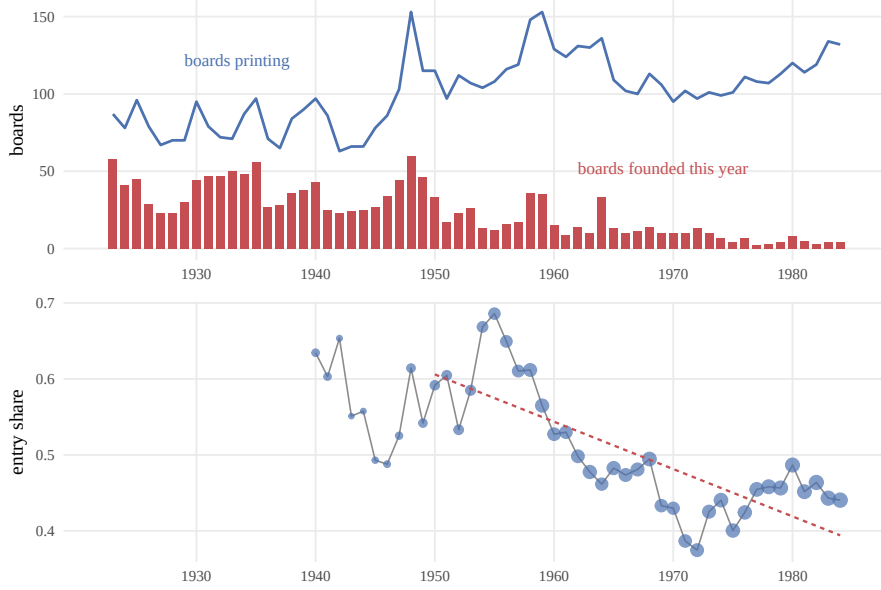


Figure 1: The field stops admitting new producing units (top) and the share of slots going to someone new falls (bottom). Top: boards founded in each year (bars) against the standing population of boards printing (line). Bottom: author slots new to the board as a share of all slots filled, on the model panel; point size is the number of slots, and the dashed line is the 1950–84 linear fit

4 Models

4.1 The exposure law

The core model for testing the neutral-null hypothesis is the Hoppe urn (Hoppe 1984). An urn holds a distinguished “mutator” ball of mass θ alongside M balls standing for the items already drawn; a draw takes the mutator, and so introduces a type not seen before, with probability

$$p = \frac{\theta}{\theta + M}. \quad (1)$$

Read as a model of a producing unit, θ is *recruitment intensity*, i.e. propensity to fetch an author from outside the catalog. M is the number of the editions already printed by the board. Equation 1 then says that the entry probability falls in M with θ held fixed. This means that an ageing unit admits proportionally fewer newcomers without having become any less inclined to admit them.

In the Hoppe urn the exponent on M is fixed at one. I add flexibility to the model by letting the exponent γ vary freely, and allow each board its own level

and each year its own effect.

$$\begin{aligned} n_{bt}^{\text{entry}} &\sim \text{Binomial}(n_{bt}, p_{bt}), \\ \text{logit } p_{bt} &= \underbrace{\alpha + \tau_t}_{\log \theta_t} + u_b - \gamma \log M_{bt}. \end{aligned} \quad (2)$$

At $\gamma = 1$ equation 2 is equation 1, since $\text{logit}\{\theta/(\theta + M)\} = \log \theta - \log M$. The urn therefore sits inside the parameter space, and γ measures how far the field departs from it. Here τ_t is a first-order random walk over years with $\tau_1 = 0$, $u_b \sim \text{Normal}(0, \sigma_u)$ is a board effect, and γ carries a weakly informative $\text{Normal}(1, 0.3)$ prior centred on the theoretical value. The model is fitted by Hamiltonian Monte Carlo in Stan, four chains of 2,000 draws. The full details of the model specification along with the fitted objects, the simulation experiment data, and various checks are available in the accompanying replication dataset (Anonymous 2026).

4.2 The estimand

The quantity of interest is the linear trend in $\log \theta_t$ over 1950–1984, computed as the ordinary least-squares slope of τ on year *inside each posterior draw*, so that uncertainty in γ propagates into it automatically and no prior asserts a sign. The decision rule is one sentence: zero or positive means exposure alone accounts for the observed decline; negative means behavioral closure on top of exposure.

5 Results

5.1 Recruitment intensity does not decline

The trend in $\log \theta_t$ over 1950–1984 is -0.0005 per year, on a 90% posterior interval of $[-0.0045, +0.0035]$, with $P(\text{trend} < 0) = 0.57$ and $P(\text{trend} < -0.01) = 0.00$. It is flat, and centred very close to zero so that its sign is not identified. Figure 2 visualizes this result by showing that recruitment intensity per unit of accumulated catalog neither fell nor rose over exactly the period in which the entry share fell from 0.59 to 0.44.

The data drive the exposure exponent γ into the sub-linear range of roughly 0.55–0.73, but the exact value is not identified well enough. The problem is that the fitted model gives an estimate of 0.56 while a fixed-effects GLM on the same likelihood gives 0.73. The $\gamma = 1$ assumed by the Hoppe urn model is rejected both ways. γ is not prior-driven: refitting under a deliberately flat $\text{Normal}(1, 1)$ moves it from 0.556 to 0.555, and the estimand from -0.0005 to -0.0006 .

5.2 Would a decline have been detected?

To claim an absence of a trend one should demonstrate that a decline, had one been present, would have shown up under a similar model. This was done in the

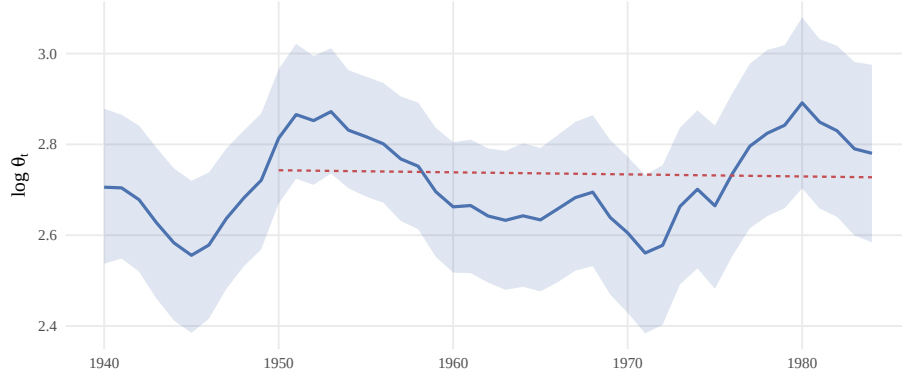


Figure 2: Recruitment intensity, the part of behavior that exposure cannot explain. Posterior mean and 90% band for $\log \theta_t$; the dashed line is the 1950–1984 estimand. Compare the falling series in the lower panel of Figure 1, over the same years and the same boards

experiment with simulated data. I planted linear declines in $\log \theta_t$, generated panels on the real design, and refitted with the same estimator — eight replicate panels in each of four conditions. Planted declines of -0.02 and -0.01 per year come back with the 90% interval entirely below zero in 8 of 8 replicates each, against 0 of 8 when nothing is planted. The recovered decline tracks the planted one one-for-one (calibration slope 0.98, $R^2 = 0.98$), and the detectable threshold is about -0.007 per year. The across-replicate spread (0.0015) matches the mean posterior standard deviation (0.0018), so the intervals are honest in width rather than merely narrow. The observed estimand sits far above the threshold.

5.3 The counterfactual

Contrary to the traditional model of cumulated advantage, the accumulated catalog value M_{bt} does not depend on the identity of the authors. It counts author *slots*, that is how many books a board printed, not who filled them. This means that there is no feedback loop between the recruitment and the exposure. As a result, the posterior can be used to create counterfactuals where θ and M are manipulated independently.

Sufficiency. Let’s freeze every board’s recruitment intensity at its 1950 value and leave its real catalog growth intact. The entry share still falls to 0.449 [0.418, 0.480] by 1984, against a fitted 0.442 and an observed 0.441. That is **96% of the fitted decline**, on a 90% interval of [72%, 124%].

Necessity. Let’s freeze the field’s exposure distribution at its 1950 shape and leave behavior free. The decline does not merely stop, it reverses, ending at 0.622 [0.604, 0.640] — above the 1950 baseline of 0.575. Only 80 of the 463 boards printing after 1950 were already printing in 1950, so this result is necessarily distributional: each board keeps its rank in the year’s $\log M$ ordering,

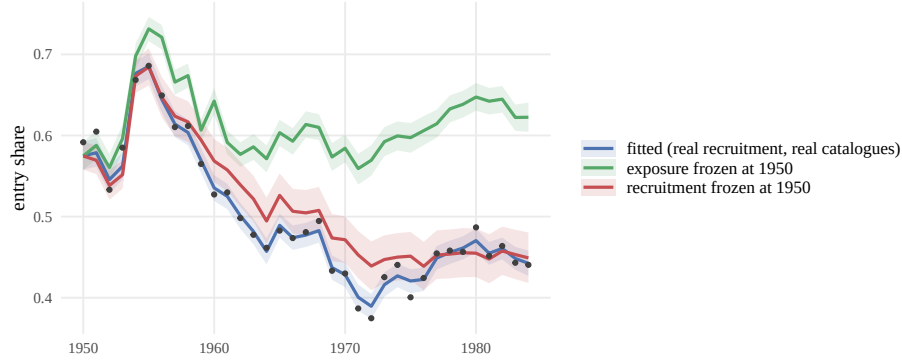


Figure 3: Exposure is sufficient, and it is necessary. Entry share implied by the fitted law under three arms, with 90% posterior bands; grey points are the observed series. All three arms coincide at the 1950 anchor by construction

and that rank is mapped onto the 1950 cross-section by the inverse empirical distribution function, which makes the anchor year reproduce the baseline exactly.

6 Discussion

The late-Soviet book publishing for children is a clear case of the ossification of the cultural field. The share of slots going to newcomers fell by a quarter between 1950 and 1984 while the propensity to recruit newcomers, measured per unit of accumulated catalog, held flat. And freezing that propensity at its 1950 value still reproduces almost the whole of the fall. So there exists an explanation of this phenomenon that does not require assumptions about the change in behavior of the gatekeepers to underlie this field level change. What did change was the age structure of the producing population. Then the assumption required for the explanation is the value of the backlog catalog for the established editorial boards, not their propensity to recruit.

Three things the estimate does not license. First, θ_t absorbs everything that changes besides the exposure law — shifts in the author pool, in cataloging recall, in what counts as an authored edition. A flat θ_t is *no evidence of change in behavior*, and not evidence of constant editorial policy. Second, the exposure exponent is weakly identified, and the two estimation routes disagree by 0.17 on the real panel against 0.06 on panels simulated from the fitted model. This can be a signature of model misspecification, most plausibly the heterogeneity in the M -slope among editorial boards. A board-varying M -slope is the specification to try next. Third, simulation from the fitted model gives 90% intervals that cover the truth 62% of the time. The two recovery biases point in opposite directions, though, +0.0037 per year high on urn-generated data and −0.0028 per year low on model-generated data, so the observed value is bracketed and

the flatness is not an artefact of a one-way bias.

The case of the Soviet book market is a proof-of-concept model that illustrates a more general idea. Wherever producing units accumulate a catalog and the population of units ages without replacement, the observed entry share is confounded by exposure. To make any judgement about the behavioral change of the institutional agents, the directly observed decline in the share of newcomers is not enough — one needs to fit θ_t . This makes institutional ageing a relevant neutral-null model, similar to a random-copying null already familiar in the study of cultural frequency data (Bentley, Hahn, and Shennan 2004; Kandler and Crema 2019). It is the neutral model to check against before any claim about disposition change is made. Read that way, rising concentration of the top-performers in a cultural field is not yet evidence about what gatekeepers prefer. It may be evidence about how old they are.

References

- Anonymous (Aug. 2026). *Replication data for: Ageing institutions ossify a cultural field: Evidence from Soviet book publishing for children*. Version 1.0. Zenodo. DOI: [10.5281/zenodo.XXXXXXX](https://doi.org/10.5281/zenodo.XXXXXXX). URL: <https://doi.org/10.5281/zenodo.XXXXXXX>.
- Bentley, R Alexander, Matthew W Hahn, and Stephen J Shennan (2004). “Random drift and culture change”. In: *Proceedings of the Royal Society B: Biological Sciences* 271.1547, pp. 1443–1450.
- Bourdieu, Pierre (1993). *The field of cultural production: Essays on art and literature*. Columbia University Press.
- DiPrete, Thomas A and Gregory M Eirich (2006). “Cumulative advantage as a mechanism for inequality: A review of theoretical and empirical developments”. In: *Annu. Rev. Sociol.* 32.1, pp. 271–297.
- Guillory, John (1993). *Cultural Capital: The Problem of Literary Canon Formation*. University of Chicago Press.
- Hannan, Michael T and John Freeman (1984). “Structural inertia and organizational change”. In: *American sociological review*, pp. 149–164.
- Hoppe, Fred M (1984). “Pólya-like urns and the Ewens’ sampling formula”. In: *Journal of Mathematical Biology* 20.1, pp. 91–94.
- Hubbell, Stephen P (2001). *The unified neutral theory of biodiversity and biogeography*. Princeton University Press.
- Kandler, Anne and Enrico R Crema (2019). “Analysing cultural frequency data: Neutral theory and beyond”. In: *Handbook of evolutionary research in archaeology*. Springer, pp. 83–108.
- Maslinsky, Kirill (2022). *Bibliografiia detskoi knigi [Bibliography of children’s books] 1918–1984*. Version V5. DOI: [10.31860/openlit-2022.12-B010](https://doi.org/10.31860/openlit-2022.12-B010). URL: <https://doi.org/10.31860/openlit-2022.12-B010>.
- Mastroianni, Adam (May 2, 2022). *Pop Culture Has Become an Oligopoly*. URL: <https://www.experimental-history.com/p/pop-culture-has-become-an-oligopoly> (visited on 08/14/2026).

- Merton, Robert K (1968). “The Matthew effect in science: The reward and communication systems of science are considered.” In: *Science* 159.3810, pp. 56–63.
- Salganik, Matthew J, Peter Sheridan Dodds, and Duncan J Watts (2006). “Experimental study of inequality and unpredictability in an artificial cultural market”. In: *science* 311.5762, pp. 854–856.
- Sørensen, Jesper B and Toby E Stuart (2000). “Aging, obsolescence, and organizational innovation”. In: *Administrative science quarterly* 45.1, pp. 81–112.